

- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Chronic Kidney Disease

CKD: Assessment: Executive Summary of Recommendations (2020)

Executive Summary of Recommendations

Below are the recommendations and ratings for the Academy of Nutrition and Dietetics 2020 Chronic Kidney Disease (CKD) Assessment Evidence-Based Nutrition Practice Guideline. Use the links on the left to view the Guideline Introduction. Detailed recommendations, including the evidence supporting these recommendations, is available from the **Major Recommendations** tab.

- For a description of the Academy Recommendation Rating scheme (Strong, Fair, Weak, Consensus, Insufficient Evidence), [click here](#).
- For a description of the GRADE Recommendation Rate scheme (Level 1 (1A, 1B, 1C, 1D), Level 2 (2A, 2B, 2C, 2D), [click here](#)

- [Screening and Referral](#)

CKD: Routine Nutrition Screening

In adults with CKD 3-5D or posttransplantation, it is reasonable to consider routine nutrition screening at least biannually with the intent of identifying those at risk of protein-energy wasting (OPINION).

**Consensus
Conditional**

CKD: Nutrition Screening Tools

In adults with CKD 3-5D or posttransplantation, there is limited evidence to suggest the use of one tool over others for identifying those at risk of protein-energy wasting (2D).

**Weak
Conditional**

- [Nutrition Assessment](#)

CKD: Routine Nutrition Assessment

In adults with CKD 3-5D or posttransplantation, it is reasonable that a registered dietitian nutritionist (RDN) or an international equivalent conduct a comprehensive nutrition assessment (including but not limited to appetite, history of dietary intake, biochemical data, anthropometric measurements, and nutrition-focused physical findings) at least within the first 90 days of starting dialysis, annually, or when indicated by nutrition screening or provider referral (OPINION).

**Consensus
Conditional**

CKD: Skinfold Thickness

In adults with CKD 1-5D (1B) or posttransplantation (OPINION), in the absence of edema, we suggest using skinfold thickness measurements to assess body fat.

**Strong
Conditional**

CKD: Waist Circumference

In adults with CKD 5D, we suggest that waist circumference may be used to assess abdominal obesity, but its reliability in assessing changes over time is low (2C).

**Weak
Conditional**

CKD: Conicity Index

In adults with CKD 5D on MHD, we suggest that the conicity index may be used to assess nutritional status (OPINION) and as a predictor of mortality (2C).

**Weak
Conditional**

CKD: Creatinine Kinetics

In adults with CKD 5D, we suggest that creatinine kinetics may be used to estimate muscle mass, though very high or very low dietary intake of meat and/or creatine supplements will influence accuracy of this measurement (2C).

**Weak
Conditional**

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Maintenance Hemodialysis (MHD)

In adults with CKD 5D on MHD, we suggest that overweight/obese status (based on BMI) can be used as a predictor of lower mortality, whereas, underweight status and morbid obesity (based on BMI) can be used as a predictor of higher mortality (2B).

**Fair
Conditional**

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Peritoneal Dialysis (PD)

In adults with CKD 5D on PD, we suggest that underweight status (based on BMI) can be used as a predictor of higher mortality (2C).

**Weak
Conditional**

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Non-Dialyzed

In adults with CKD 1-5, it is reasonable to consider using underweight status (based on BMI) as a predictor of higher mortality, though the mortality risk associated with overweight or obesity status (based on BMI) is not clear (OPINION).

**Consensus
Conditional**

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Post-Transplant

In posttransplantation adults, it is reasonable to consider using underweight and overweight/obesity status (based on BMI) as a predictor of higher mortality (OPINION).

**Consensus
Conditional**

CKD: Body Composition and Body Weight/Body Mass Index (BMI)

In adults with CKD 1-5D or posttransplantation, it is reasonable to consider assessing body composition in combination with body weight/BMI at the first visit and to monitor overall nutrition status periodically over time (OPINION).

**Consensus
Conditional**

CKD: Frequency of Body Weight/Body Mass Index (BMI) and Body Composition Assessment

In adults with CKD 1-5D or posttransplantation who are clinically stable, it is reasonable to measure body weight and BMI and to monitor for changes in body weight/BMI and body composition as needed (OPINION).

- At least monthly in MHD and PD patients
- At least every 3 months in patients with CKD 4-5 or posttransplantation
- At least every 6 months in patients with CKD 1-3

**Consensus
Conditional****CKD: Assessment of Body Weight**

In adults with CKD 1-5D or posttransplantation, it is reasonable for a registered dietitian nutritionist (RDN) or an international equivalent or physicians to use clinical judgement to determine the method for measuring body weight (e.g. actual measured weight, history of weight changes, serial weight measurements, adjustments for suspected impact of edema, ascites and polycystic organs) due to absence of standard reference norms (OPINION).

**Consensus
Conditional****CKD: Body Mass Index (BMI) and Protein Energy Wasting (PEW)**

In adults with CKD 1-5D or posttransplantation, BMI alone is not sufficient to establish a diagnosis of PEW unless the BMI is very low (<18 kg/m²) (OPINION).

**Consensus
Conditional****CKD: Bioelectrical Impedance for Patients on Maintenance Hemodialysis (MHD)**

In adults with CKD 5D on MHD, we suggest using bioimpedance and preferably multi-frequency bioelectrical impedance (MF-BIA) to assess body composition when available. Bioimpedance assessments should ideally be performed a minimum of 30 minutes or more after the end of the hemodialysis session to allow for redistribution of body fluids (2C).

**Weak
Conditional****CKD: Bioelectrical Impedance for Patients, Non-Dialyzed and on Peritoneal Dialysis (PD)**

In adults with CKD 1-5 or CKD 5D on PD, there is insufficient evidence to suggest using bioelectrical impedance to assess body composition (2D).

**Weak
Conditional****CKD: Dual-Energy X-Ray Absorptiometry (DXA) for Body Composition Assessment**

In adults with CKD 1-5D or posttransplantation, it is reasonable to use DXA when feasible as it remains the gold standard for measuring body composition despite being influenced by volume status (OPINION).

**Consensus
Conditional****CKD: Single Biomarker Measurements**

In adults with CKD stages 1-5D or posttransplantation, biomarkers such as normalized protein catabolic rate (nPCR), serum albumin and/or serum prealbumin (if available) may be considered complementary tools to assess nutritional status. However, they should not be interpreted in isolation to assess nutritional status as they are influenced by non-nutritional factors (OPINION).

**Consensus
Conditional****CKD: Serum Albumin Levels**

In adults with CKD 5D on MHD, serum albumin may be used as a predictor of hospitalization and mortality, with lower levels associated with higher risk (1A).

**Strong
Imperative****CKD: Handgrip Strength**

In adults with CKD 1-5D, we suggest that handgrip strength may be used as an indicator of protein-energy status and functional status when baseline data (prior measures) are available for comparison (2B).

**Fair
Conditional****CKD: Assessment of Resting Energy Expenditure**

In adults with CKD 1-5D or posttransplantation, it is reasonable to use indirect calorimetry to measure resting energy expenditure when feasible and indicated, as it remains the gold standard for determining resting energy expenditure (OPINION).

**Consensus
Conditional****CKD: Resting Energy Expenditure Equations**

In adults with CKD 5D who are metabolically stable, we suggest that in the absence of indirect calorimetry, disease-specific predictive energy equations may be used to estimate resting energy expenditure as they include factors that may influence the metabolic rate in this population (2C).

**Weak
Conditional****CKD: 7-Point Subjective Global Assessment (SGA)**

In adults with CKD 5D, we recommend the use of the 7-point Subjective Global Assessment as a valid and reliable tool for assessing nutritional status (1B).

**Strong
Imperative****CKD: Malnutrition Inflammation Score (MIS)**

In adults with CKD 5D on MHD or posttransplantation, Malnutrition Inflammation Score may be used to assess nutritional status (2C).

**Weak
Conditional****CKD: Considerations when Assessing Dietary Intake**

In adults with CKD 3-5D or posttransplantation, it is reasonable to assess factors beyond dietary intake (e.g. medication use, knowledge, beliefs, attitudes, behavior and access to food, depression, cognitive function etc.) to effectively plan nutrition interventions. (OPINION).

**Consensus
Conditional****CKD: 3-Day Food Records to Assess Dietary Intake**

In adults with CKD 3-5D, we suggest the use of a 3-day food record, conducted during both dialysis and non-dialysis treatment days (when applicable), as a preferred method to assess dietary intake (2C).

Weak
Conditional

CKD: Alternative Methods of Assessing Dietary Intake in CKD 3-5

In adults with CKD 3-5 (OPINION), or CKD 5D (2D), 24-hour food recalls, food frequency questionnaires and normalized protein catabolic rate (nPCR) may be considered as alternative methods of assessing dietary energy and protein intake (2D).

Consensus
Conditional

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Chronic Kidney Disease

CKD: Macronutrients: Executive Summary of Recommendations (2020)

Executive Summary of Recommendations

Below are the recommendations and ratings for the Academy of Nutrition and Dietetics 2020 Chronic Kidney Disease (CKD) Macronutrient Evidence-Based Nutrition Practice Guideline. Use the links on the left to view the Guideline Introduction. Detailed recommendations, including the evidence supporting these recommendations, is available from the **Major Recommendations** tab.

- For a description of the Academy Recommendation Rating scheme (Strong, Fair, Weak, Consensus, Insufficient Evidence), [click here](#).
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• [Nutrition Intervention](#)

CKD: Dietary Patterns: Mediterranean Diet

In adults with CKD 1-5 not on dialysis or posttransplantation, with or without dyslipidemia, we suggest that prescribing a Mediterranean Diet may improve lipid profiles (2C).

Weak
Conditional

CKD: Dietary Patterns: Fruits and Vegetables

In adults with CKD 1-4, we suggest that prescribing increased fruit and vegetable intake may decrease body weight, blood pressure and net acid production (NEAP) (2C).

Weak
Conditional

CKD: LC n-3 PUFA Nutritional Supplements for Lipid Profile, Maintenance Hemodialysis

In adults with CKD 5D on MHD, we suggest that 1.3-4 g/d LC n-3 PUFA may be prescribed to reduce triglycerides and LDL cholesterol (2C) and raise HDL levels (2D).

Weak
Conditional

CKD: LC n-3 PUFA Nutritional Supplements for Lipid Profile, Peritoneal Dialysis

In adults with CKD 5D on PD, it is reasonable to consider prescribing 1.3-4 g/d LC n-3 PUFA to improve the lipid profile (OPINION).

Consensus
Conditional

CKD: LC n-3 PUFA Nutritional Supplements for Lipid Profile, Non-dialyzed

In adults with CKD 3-5, we suggest prescribing ~2g/d LC n-3 PUFA to lower serum triglyceride levels (2C).

Weak
Conditional

CKD: LC n-3 PUFA Nutritional Supplements for Mortality and Cardiovascular Disease, Maintenance Hemodialysis and Post-Transplant

In adults with CKD 5D on MHD or posttransplantation, we suggest not routinely prescribing LC n-3 PUFA, including those derived from fish or flaxseed and other oils, to lower risk of mortality (2C) or cardiovascular events (2B).

Fair
Imperative

CKD: LC n-3 PUFA Nutritional Supplements for Mortality and Cardiovascular Disease, Peritoneal Dialysis

In adults with CKD 5D on PD, it is reasonable to not routinely prescribe LC n-3 PUFA, including those derived from fish or flaxseed and other oils, to lower risk of mortality or cardiovascular events (OPINION).

Consensus
Conditional

CKD: Protein Restriction, Non-Dialysis, Non-Diabetic

In adults with CKD 3-5 who are metabolically stable, we recommend, under close clinical supervision, protein restriction with or without keto acid analogs, to reduce risk for ESKD/death (1A) and improve quality of life (2C).

- a low protein diet providing 0.55 to 0.60 g dietary protein/kg body weight/day , OR
- a very-low protein diet providing 0.28 to 0.43 g dietary protein/kg body weight/day with additional keto acid/amino acid analogs to meet protein requirements (0.55 to 0.60 g /kg body weight/day)

Strong
Conditional

CKD: Dietary Protein Intake, Maintenance Hemodialysis, Non-Diabetic

In adults with CKD 5D on MHD (1C) who are metabolically stable, we recommend prescribing a dietary protein intake of 1.0 -1.2 g /kg body weight per day to maintain a stable nutritional status.

Fair
Conditional

CKD: Dietary Protein Intake, Peritoneal Dialysis, Non-Diabetic

In adults with CKD 5D on PD (OPINION) who are metabolically stable, we recommend prescribing a dietary protein intake of 1.0 -1.2 g /kg body weight per day to maintain a stable nutritional status.

**Consensus
Conditional**

CKD: Macronutrients: Protein Type

In adults with CKD 1-5D, there is insufficient evidence to recommend a particular protein type (plant vs animal) in terms of the effects on nutritional status, calcium or phosphorus levels, or the blood lipid profile (1B).

**Strong
Imperative**

CKD: Macronutrients: Protein Type, Post-Transplant

In adults with CKD posttransplantation, there is insufficient evidence to recommend a particular protein type (plant vs animal) in terms of the effects on nutritional status, calcium or phosphorus levels, or the blood lipid profile (OPINION).

**Consensus
Conditional**

CKD: LC n-3 PUFA Nutritional Supplements for AV Graft and Fistula Patency for Maintenance Hemodialysis

In adults with CKD 5D on maintenance hemodialysis (MHD), we suggest not routinely prescribing fish oil to improve primary patency rates in patients with AV grafts (2B) or fistulas (2A).

**Fair
Imperative**

CKD: LC n-3 PUFA Nutritional Supplements for Kidney Allograft Survival

In posttransplantation adults, we suggest not routinely prescribing LC n-3 PUFA to reduce the number of rejection episodes or improve graft survival (2D).

**Weak
Conditional**

CKD: Oral Protein-Energy Supplementation for CKD 3-5D

In adults with CKD 3-5D at risk of or with protein-energy wasting, we suggest a minimum of a 3-month trial of oral nutritional supplements to improve nutritional status if dietary counselling alone does not achieve sufficient energy and protein intake to meet nutritional requirements (2D).

**Weak
Conditional**

CKD: Oral Protein-Energy Supplementation for Post-Transplant

In adults with CKD posttransplantation at risk of or with protein-energy wasting, it is reasonable to consider a minimum of a 3-month trial of oral nutritional supplements to improve nutritional status if dietary counselling alone does not achieve sufficient energy and protein intake to meet nutritional requirements (OPINION).

**Consensus
Conditional**

CKD: Enteral Nutrition Supplementation

In adults with CKD 1-5D, with chronically inadequate intake and whose protein and energy requirements cannot be attained by dietary counselling and oral nutritional supplements, it is reasonable to consider a trial of enteral tube feeding (OPINION).

**Consensus
Conditional**

CKD: Total Parenteral Nutrition (TPN) and Intradialytic Parenteral Nutrition (IDPN) Protein-Energy Supplementation

In adults with CKD with protein-energy wasting, we suggest a trial of total parenteral nutrition (TPN) for CKD 1-5 patients and IDPN for CKD 5D on MHD patients, to improve and maintain nutritional status if nutritional requirements cannot be met with existing oral and enteral intake (2C).

**Weak
Conditional**

CKD: Dialysate Protein-Energy Supplementation

In adults with CKD 5D on PD with protein-energy wasting, we suggest not substituting conventional dextrose dialysate with amino acid dialysate as a general strategy to improve nutritional status, although it is reasonable to consider a trial of amino acid dialysate to improve and maintain nutritional status if nutritional requirements cannot be met with existing oral and enteral intake (OPINION).

**Consensus
Conditional**

CKD: Protein Restriction, Non-Dialysis, Diabetic

In adults with CKD 3-5 and who have diabetes, it is reasonable to prescribe, under close supervision, a dietary protein intake of 0.6–0.8 g /kg body weight per day to maintain a stable nutritional status and optimize glycemic control (OPINION).

**Consensus
Conditional**

CKD: Dietary Protein Intake, Maintenance Hemodialysis and Peritoneal Dialysis, Diabetic

In adults with CKD 5D and who have diabetes, it is reasonable to prescribe a dietary protein intake of 1.0 -1.2 g /kg body weight per day to maintain a stable nutritional status. For patients at risk of hyper and/or hypoglycemia, higher levels of dietary protein intake may need to be considered to maintain glycemic control (OPINION).

**Consensus
Conditional**

CKD: Statement on Energy Intake in CKD 1-5D

In adults with CKD 1-5D (1C) and who are metabolically stable, we recommend prescribing an energy intake of 25-35 kcal/kg body weight per day based on age, gender, level of physical activity, body composition, weight status goals, CKD stage, and concurrent illness or presence of inflammation to maintain normal nutritional status.

**Fair
Imperative**

CKD: Statement on Energy Intake in Post-Transplant

In adults with CKD posttransplantation (OPINION) who are metabolically stable, we recommend prescribing an energy intake of 25-35 kcal/kg body weight per day based on age, gender, level of physical activity, body composition, weight status goals, CKD stage, and concurrent illness or presence of inflammation to maintain normal nutritional status.

**Consensus
Conditional**

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Chronic Kidney Disease

CKD: Micronutrient: Executive Summary of Recommendations (2020)

Executive Summary of Recommendations

Below are the recommendations and ratings for the Academy of Nutrition and Dietetics 2020 Chronic Kidney Disease (CKD) Micronutrient Evidence-Based Nutrition Practice Guideline. Use the links on the left to view the Guideline Introduction. Detailed recommendations, including the evidence supporting these recommendations, is available from the **Major Recommendations** tab.

- For a description of the Academy Recommendation Rating scheme (Strong, Fair, Weak, Consensus, Insufficient Evidence), [click here](#).
- For a description of the GRADE Recommendation Rate scheme (Level 1 (1A, 1B, 1C, 1D), Level 2 (2A, 2B, 2C, 2D), [click here](#)

• [Nutrition Intervention](#)

CKD: Dietary Micronutrient Intake

In adults with CKD 3-5D or posttransplantation, it is reasonable for the registered dietitian nutritionist (RDN) or an international equivalent, to encourage eating a diet that meets the recommended dietary allowance (RDA) for adequate intake for all vitamins and minerals (OPINION)

**Consensus
Conditional**

CKD: Micronutrient Assessment and Supplementation

In adults with CKD 3-5D or posttransplantation, it is reasonable for the registered dietitian nutritionist (RDN) or an international equivalent, in close collaboration with a physician or physician assistant, to assess dietary vitamin intake periodically and to consider multivitamin supplementation for individuals with inadequate vitamin intake (OPINION).

**Consensus
Conditional**

CKD: Micronutrient Supplementation, Dialysis

In adults with CKD 5D who exhibit inadequate dietary intake for sustained periods of time, it is reasonable to consider supplementation with multivitamins, including all the water-soluble vitamins, and essential trace elements to prevent or treat micronutrient deficiencies (OPINION).

**Consensus
Conditional**

CKD: Folic Acid Supplementation for Hyperhomocysteinemia

In adults with CKD 3-5D or posttransplantation who have hyperhomocysteinemia associated with kidney disease, we recommend not to routinely supplementing folate with or without B-complex since there is no evidence demonstrating reduction in adverse cardiovascular outcomes (1A).

**Strong
Conditional**

CKD: Folic Acid Supplementation for Folic Acid Deficiency and Insufficiency, CKD 1-5 and Dialysis

In adults with CKD 1-5D, we suggest prescribing folate, Vitamin B12 and/or B-complex supplement to correct for folate or Vitamin B12 deficiency/insufficiency, based on clinical signs and symptoms (B2).

**Fair
Conditional**

CKD: Folic Acid Supplementation for Folic Acid Deficiency and Insufficiency, CKD Post-Transplant

In adults with CKD posttransplantation, we suggest prescribing folate, Vitamin B12 and/or B-complex supplement to correct for folate or Vitamin B12 deficiency/insufficiency based on clinical signs and symptoms (OPINION).

**Consensus
Conditional**

CKD: Vitamin C Supplementation

In adults with CKD 1-5D or posttransplantation who are at risk of Vitamin C deficiency it is reasonable to consider supplementation to meet the recommended intake of at least 90 mg/d for men and 75 mg/d for women (OPINION).

**Consensus
Conditional**

CKD: Vitamins E and A Supplementation and Toxicity

In adults with CKD 5D on MHD or CKD 5D on PD, it is reasonable to not routinely supplement Vitamin A or E because of the potential for vitamin toxicity. However, if supplementation is warranted, care should be taken to avoid excessive doses, and patients should be monitored for toxicity (OPINION).

**Consensus
Imperative**

CKD: Anticoagulant Medication and Vitamin K Supplementation

In adults with CKD 1-5D or posttransplantation, it is reasonable that patients receiving anticoagulant medicines known to inhibit vitamin K activity (e.g., warfarin compounds) do not receive vitamin K supplements (OPINION).

**Consensus
Conditional**

CKD: Selenium and Zinc Supplementation

In adults with CKD 1-5D, we suggest to not routinely supplementing selenium or zinc since there is little evidence that it improves nutritional, inflammatory or micronutrient status (2C).

**Weak
Imperative**

CKD: Vitamin D Supplementation for Vitamin D Deficiency and Insufficiency, CKD 1-5D

In adults with CKD 1-5D (2C), we suggest prescribing Vitamin D supplementation in the form of cholecalciferol or ergocalciferol to correct 25(OH)D deficiency/insufficiency.

**Weak
Conditional**

CKD: Vitamin D Supplementation for Vitamin D Deficiency and Insufficiency, CKD Post-Transplant

In adults with CKD posttransplantation, we suggest prescribing Vitamin D supplementation in the form of cholecalciferol or ergocalciferol to correct 25(OH)D deficiency/insufficiency (OPINION).

**Consensus
Conditional**

CKD: Vitamin D Supplementation with Proteinuria

In adults with CKD 1-5 with nephrotic-range proteinuria, it is reasonable to consider supplementation of cholecalciferol, ergocalciferol or other safe and effective 25(OH)D precursors (OPINION).

**Consensus
Conditional**

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Chronic Kidney Disease

CKD: Electrolytes and Other Nutrients: Executive Summary of Recommendations (2020)

Executive Summary of Recommendations

Below are the recommendations and ratings for the Academy of Nutrition and Dietetics 2020 Chronic Kidney Disease (CKD) Electrolyte and Other Nutrients Evidence-Based Nutrition Practice Guideline. Use the links on the left to view the Guideline Introduction. Detailed recommendations, including the evidence supporting these recommendations, is available from the **Major Recommendations** tab.

- For a description of the Academy Recommendation Rating scheme (Strong, Fair, Weak, Consensus, Insufficient Evidence), [click here](#).
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- [Nutrition Intervention](#)

CKD: Dietary Phosphorus Amount

In adults with CKD 3-5D, we recommend adjusting dietary phosphorus intake to maintain serum phosphate levels in the normal range (1B).

**Strong
Imperative**

CKD: Dietary Phosphorus Source

In adults with CKD 1-5D or posttransplantation, it is reasonable when making decisions about phosphorus restriction treatment to consider the bioavailability of phosphorus sources (e.g. animal, vegetable, additives) (OPINION).

**Consensus
Conditional**

CKD: Phosphorus Intake with Hypophosphatemia

For posttransplantation adults with hypophosphatemia, it is reasonable to consider prescribing high-phosphorus intake (diet or supplements) in order to replete serum phosphate (OPINION).

**Consensus
Conditional**

CKD: Sodium Intake and Blood Pressure, CKD 3-5, Non-Dialyzed

In adults with CKD 3-5 not on dialysis, we recommend limiting sodium intake to less than 100 mmol/day (or <2.3 g/day) to reduce blood pressure and improve volume control (1B).

**Strong
Imperative**

CKD: Sodium Intake and Blood Pressure, CKD 5D and Post-Transplant

In adults with CKD 5D or posttransplantation, we recommend limiting sodium intake to less than 100 mmol/day (or <2.3 g/day) to reduce blood pressure and improve volume control (1C).

**Fair
Imperative**

CKD: Sodium Intake and Proteinuria

In adults with CKD 3-5, we suggest limiting sodium intake to less than 100 mmol/day (or <2.3 g/day) to reduce proteinuria synergistically with available pharmacological intervention (2A).

**Fair
Conditional**

CKD: Sodium Intake and Dry Body Weight

In adults with CKD 3-5D, we suggest reduced dietary sodium intake as an adjunctive lifestyle modification strategy to achieve better volume control and a more desirable body weight (2B).

**Fair
Conditional**

CKD: Dietary Potassium Amount

In adults with CKD 3-5D or posttransplantation, it is reasonable to adjust dietary potassium intake to maintain serum potassium within the normal range (OPINION).

**Consensus
Conditional**

CKD: Dietary and Supplemental Potassium Intake for Hyperkalemia or Hypokalemia, CKD Post-Transplant

In adults with CKD posttransplantation with either hyperkalemia or hypokalemia, we suggest that dietary or supplemental potassium intake be based on patient's individual needs and clinician judgement (OPINION).

**Consensus
Conditional**

CKD: Dietary and Supplemental Potassium Intake for Hyperkalemia or Hypokalemia, CKD 3-5D

In adults with CKD 3-5D with either hyperkalemia or hypokalemia, we suggest that dietary or supplemental potassium intake be based on a patient's individual needs and clinician judgment (2D).

**Weak
Imperative**

CKD: Dietary Management of Net Acid Production (NEAP)

In adults with CKD 1-4, we suggest reducing net acid production (NEAP) through increased dietary intake of fruits and vegetables (2C) in order to reduce the rate of decline of residual kidney function.

**Weak
Conditional**

CKD: Bicarbonate Supplementation

In adults with CKD 3-5D, we recommend reducing net acid production (NEAP) through increased bicarbonate or a citric acid/sodium citrate solution supplementation (1C) in order to reduce the rate of decline of residual kidney function.

**Fair
Conditional**

CKD: Bicarbonate Maintenance

In adults with CKD 3-5D, it is reasonable to maintain serum bicarbonate levels at 24-26 mmol/L (OPINION).

**Consensus
Conditional**

CKD: Total Calcium Intake, CKD 3-4

In adults with CKD 3-4 not taking active Vitamin D analogs, we suggest that a total elemental calcium intake of 800-1,000 mg/d (including dietary calcium, calcium supplementation and calcium-based phosphate binders) be prescribed to maintain a neutral calcium balance (2B).

**Fair
Conditional**

CKD: Total Calcium Intake, CKD 5D

In adults with CKD 5D, it is reasonable to adjust calcium intake (dietary calcium, calcium supplements or calcium-based binders) with consideration of concurrent use of vitamin D analogs and calcimimetics in order to avoid hypercalcemia or calcium overload (OPINION).

**Consensus
Conditional**

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Chronic Kidney Disease

CKD: Medical Nutrition Therapy: Executive Summary of Recommendations (2020)

Executive Summary of Recommendations

Below are the recommendations and ratings for the Academy of Nutrition and Dietetics 2020 Chronic Kidney Disease (CKD) Medical Nutrition Therapy Evidence-Based Nutrition Practice Guideline. Use the links on the left to view the Guideline Introduction. Detailed recommendations, including the evidence supporting these recommendations, is available from the **Major Recommendations** tab.

- For a description of the Academy Recommendation Rating scheme (Strong, Fair, Weak, Consensus, Insufficient Evidence), [click here](#).
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- [Nutrition Intervention](#)

CKD: MNT to Improve Outcomes

In adults with CKD 1-5D, we recommend that a registered dietitian nutritionist (RDN) or an international equivalent, in close collaboration with a physician, or other provider (nurse practitioner or physician assistant), provide medical nutrition therapy (MNT). Goals are to optimize nutritional status, and to minimize risks imposed by co-morbidities and alterations in metabolism on the progression of kidney disease (1C) and on adverse clinical outcomes (OPINION).

**Fair
Imperative**

CKD: MNT Content

In adults with CKD 1-5D or posttransplantation, it is reasonable to prescribe MNT that is tailored to the individuals' needs, nutritional status and co-morbid conditions (OPINION).

**Consensus
Imperative**

- [Nutrition Monitoring and Evaluation](#)

CKD: MNT Monitoring and Evaluation

In adults with CKD 3-5D or posttransplantation, it is reasonable for the registered dietitian nutritionist (RDN) or an international equivalent to monitor and evaluate appetite, dietary intake, body weight changes, biochemical data, anthropometric measurements, and nutrition-focused physical findings to assess the effectiveness of medical nutrition therapy (OPINION).

**Consensus
Imperative**

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